Simpson's Paradox — Corpus Summary

A digital mirror of Simpson's Paradox — the paradox falls out of the DAG as a derived fact, never modeled directly.

Generated 2026-09-06 — a status report on a live, running instrument, not a one-off analysis. Every number below is queried fresh from Postgres views computed straight off the rulebook below; nothing here is hand-typed.

The single source of truth for everything in this report — jump in any time:
effortless-rulebook/simpsons-paradox-rulebook.json

What is this?

Digital mirror of the Simpson's Paradox domain. The entities are Studies, Treatments, Strata, and CaseCells (raw patient counts). SuccessRates, TreatmentSummaries, and TreatmentRankings are derived. The paradox falls out of the DAG as an emergent derived fact — it is never modeled directly.

Domain mirror of Simpson's Paradox. Entities: Studies, Treatments, Strata, CaseCells. CellSuccessRate is the first derived fact. Pooled rates, per-stratum rates, and the reversal fall out as higher-order derived nodes in subsequent loops.

What it is for:

- Teaching Simpson's Paradox with real data
- Classifying studies by distortion geometry
- Generating researcher action (PolicyImplication) from allocation bias
- Prospective literature screening for confounded pooled conclusions

The instrument witnessed its first full type-A reversal from real published data (reintjes-2000 and radelet-1981) in loop-20b. Berkeley (loop-08) introduced the partial reversal and the sign-flip definition replaced the unanimity criterion in loop-17. The four-type taxonomy (A/B/C/D) was completed in loop-21.

How to read this report

This rulebook is a high-fidelity measurement instrument, not an oracle of truth. It classifies allocation geometry precisely and repeatably — sign flips, distortion ratios, signal purity — the same way a thermometer measures temperature. It does not adjudicate causal truth. Causal interpretation is external, gated by AdjustmentAppropriate, and rests on CausalRole, the one hand/LLM-encoded field in the DAG. Berkeley: highest AllocationDistortion (0.193) but CausalRole=contested — geometry and confounding are not the same thing, and the instrument only measures the former.

- Theorem conclusions follow from definitions (regression checks for the transpiler). Domain conclusions and corpus discovery findings are statistics on a curated convenience sample — directional, not inferential. See conc-14 before treating cross-domain patterns as laws.
- Allocation sweep: 10 steps per study, f ∈ [0.05, 0.95] (SweepStudyConfig). Latent flip = Type D at observed allocation + PooledGapCrossesZero somewhere in that range — not necessarily at observed allocation.
- Most "InvariantChecks" rows are transpiler regression checks restating one algebraic fact in a second vocabulary (definitional) — not independent empirical tests. A few cross into the one hand/LLM-encoded field (CausalRole, judgment-crossing) or are currently empty-universe (vacuous). See "Invariant health" below for the breakdown.

Epistemic summary (tiered)

Proved by construction (pure algebra — no CausalRole dependency): 4
Proved, conditional on CausalRole annotation (see caveats below — not pure algebra): 4
Established in instrument (instrument · taxonomy · methodology · scope): 24
Observed in this corpus (domain — provisional): 2
Total conclusions: 34
Corpus hypotheses (loop-61): 23 / 25 confirmed
Consistency checks (definition-linked): 9 / 9 pass

Latent Type-D — flips somewhere across the full f ∈ [0.05, 0.95] sweep, not necessarily near observed allocation: 112 / 138 (81.2%)

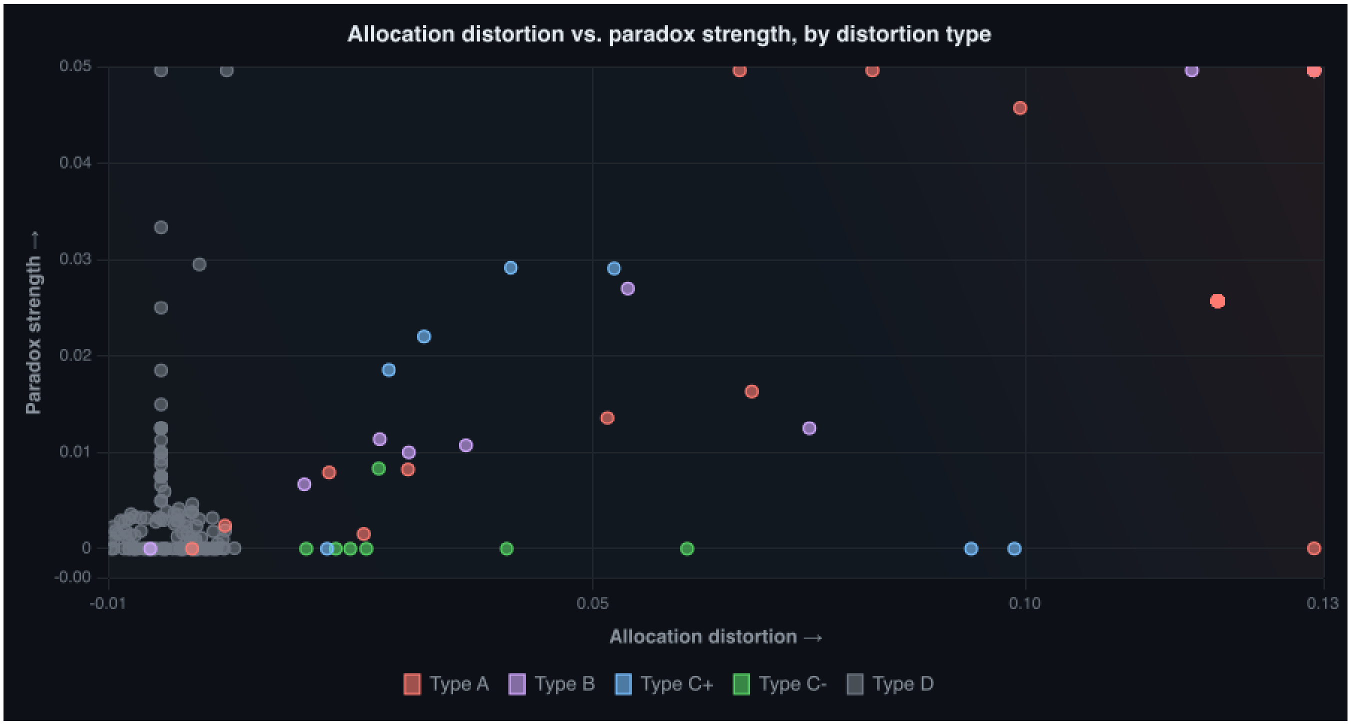
- This measures mathematical fragility over a wide counterfactual range, not practical vulnerability under realistic reallocations — see conc-15.

Studies with a full paired-treatment encoding (StudyCount, current live N — the denominator for every ratio in this report): 242

- RealStudyCount=290 counts a larger superset: every non-synthetic Studies row, including 48 control studies (loop-88) that exist as rows but were never encoded with paired treatment arms and so produce zero TreatmentRankings — not a data error, but a different denominator. Do not treat the two counts as interchangeable.

Distortion taxonomy: A 79 · B 10 · C+ 8 · C− 7 · D 138

The corpus at a glance



Proved conclusions (10)

Theorems and scope-defining findings, re-verified live against the current 242-study corpus. Evidence text below may still cite the corpus size at the loop where each theorem was first witnessed (e.g. "90+ studies," "96-study corpus") — that is historical provenance, not a claim about today's N. The pass/fail counts in "Invariant health" reflect the live corpus.

conc-09-causal-vs-geometric · Scope / open question

AllocationDistortion is a geometric measure; causal interpretation requires external domain knowledge, gated by AdjustmentAppropriate

Berkeley has the highest AllocationDistortion (0.1934) but CausalRole=contested (mediator vs confounder), proving that geometric severity and causal confounding are not the same thing. The instrument is deliberately geometric: it classifies distortion type and flags the corrected winner from allocation arithmetic alone. Whether it is safe to act on that correction as a causal claim is answered by AdjustmentAppropriate — which gates on ConditioningRisk and CausalClaimStatus. The two layers are complementary, not ...

Loop replay: loop-30
Protecting invariants: 1

conc-12-signal-purity-theorem · Proved (by construction)

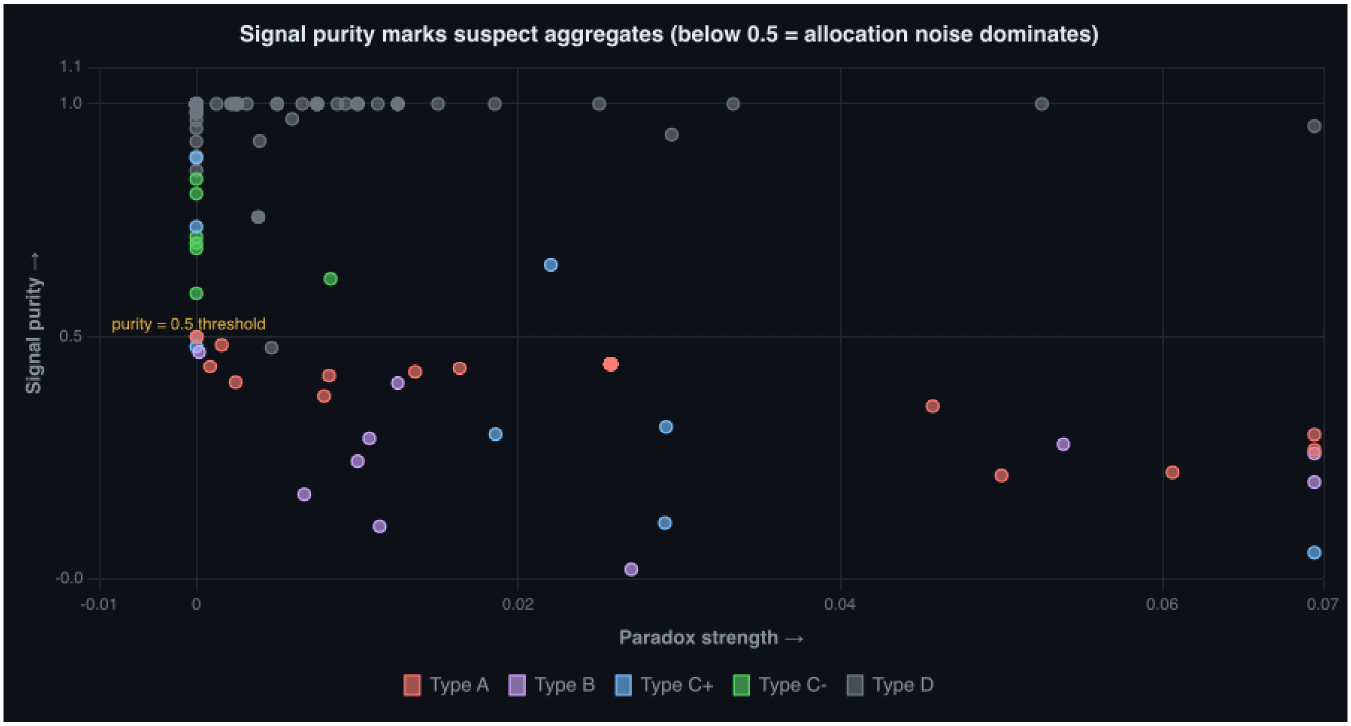
SignalPurity < 0.5 is a necessary condition for allocation-driven reversal

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

Witnessed across all reversal studies (type A + B) in the 90+ study corpus: every study with AllocationDirection='reversal' has SignalPurity < 0.5. The algebraic proof: reversal requires SignedPooledGap and CorrectedGap to have opposite signs, which requires AllocationDistortion > |CorrectedGap|, which is exactly SignalPurity < 0.5. inv-signal-purity-sign-flip enforces this as a DAG invariant. AvgSignalPurityReversal ≈ 0.35 vs AvgSignalPurityNonReversal ≈ 0.65 — gap of ~0.30.

Loop replay: [loop-44]

Protecting invariants: 1



conc-13-cplus-cminus-opposite-errors · Taxonomy

C+ (amplification) and C- (compression) are epistemically opposite errors with opposite clinical implications

titanic-1912 (C+, DistortionRatio=1.359): allocation inflates first-class survival advantage from 27.5pp corrected to 37.4pp pooled — overconfidence. melanoma-altman-1991 (C+, DistortionRatio=2.087): allocation doubles apparent female survival advantage — severe overconfidence. hannan-1994 (C-, DistortionRatio=0.314): allocation compresses CABG advantage from 8.9pp corrected to 2.8pp pooled — underconfidence, risk of premature treatment abandonment. The prior type-C category hid this distinction. Now separated b...

Loop replay: [loop-43]

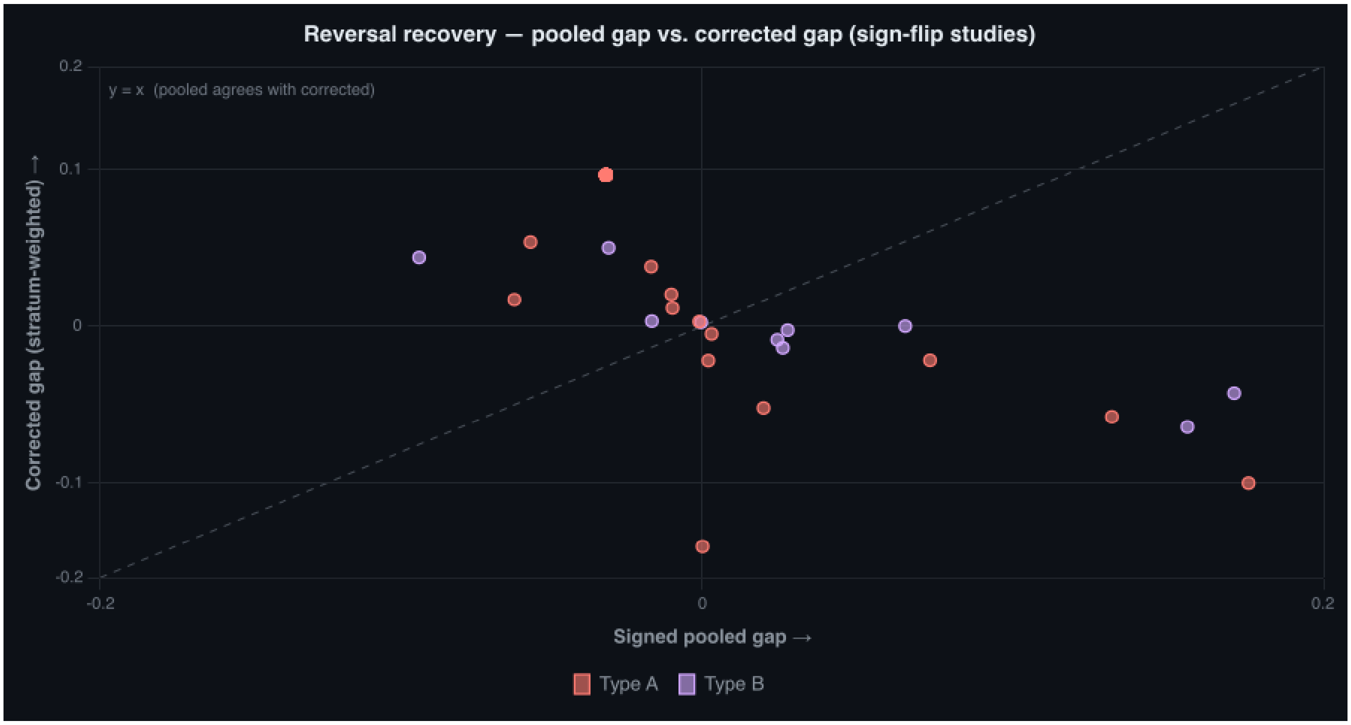
conc-20-signal-purity-ceiling · Proved (by construction)

SignalPurity < 0.5 is a build-breaking invariant for sign-flip studies — witnessed at corpus scale

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

At 96-study corpus scale, every manifest sign-flip has SignalPurity strictly below 0.5 (corpus max: 0.483, jeter-justice-1997). The melanoma-altman-1991 boundary case (purity=0.479, C+, pooled gap crosses zero, no flip) confirms the law is tight but not violated. This graduates from corpus hypothesis (loop-61) to build-breaking invariant: any future study encoding that produces a sign-flip with purity >= 0.5 is an encoding error, not a counterexample.

Loop replay: [loop-64]



conc-28-corrected-gap-invariance-theorem · Proved (by construction)

CorrectedGap (= WeightedStratumGapSum) is invariant under allocation reweighting — the conserved quantity

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

Loop-73: inv-corrected-gap-invariant PASS 238/238; MaxStudySweepCorrectedGapRange=0; CorrectedGapInvariantFailCount=0. CorrectedGap (=WeightedStratumGapSum) invariant under allocation reweighting across full corpus including expansion wave 3.

Loop replay: [loop-73]
Protecting invariants: 1

conc-29-explained-confounder-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

IsParadoxExplained ⟺ confounder manifest sign-flip — adjust-resolvable confounding is the unique explanation condition

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-74: H-explained-bidirectional PASS. ExplainedConfounderCount=ConfounderSignFlipCount; FalsePositiveExplainedCount=0; UnexplainedConfounderSignFlipCount=0. inv-explained-sign-flip-confounder + inv-confounder-signflip-explained both PASS.

Loop replay: [loop-74]
Protecting invariants: 2

conc-30-collider-no-manifest-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Given correct causal-role annotation, collider/selection studies produce no manifest sign-flips at observed allocation

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-75: ColliderSelectionCount>=10 with ColliderSelectionManifestCount=0 after encoding heckman-1979 + hoxby-murarka-2009 and re-tagging simonsen/hammond to selection; fryer-2016 reclassified contested. inv-collider-no-manifest PASS.

Loop replay: loop-75
Protecting invariants: 1

conc-35-unexplained-identity-sink-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Identity-conditioned explained theorem: all unexplained sign-flips confined to contested-org-choice and collider-proxy identities

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-85 at N=238: H-unexplained-flips-only-nonconfounders PASS. UnexplainedConfounderSignFlipCount=0. The three unexplained flips (conc-29 exception list) map exclusively to id-contested-org-choice (2) and id-collider-proxy (1). This extends the loop-74 biconditional (conc-29): explained↔confounder holds for every unambiguous confounder identity; violations are confined to ontologically ambiguous categories. inv-unexplained-identity-sink: PASS (critical, 30 total invariants).

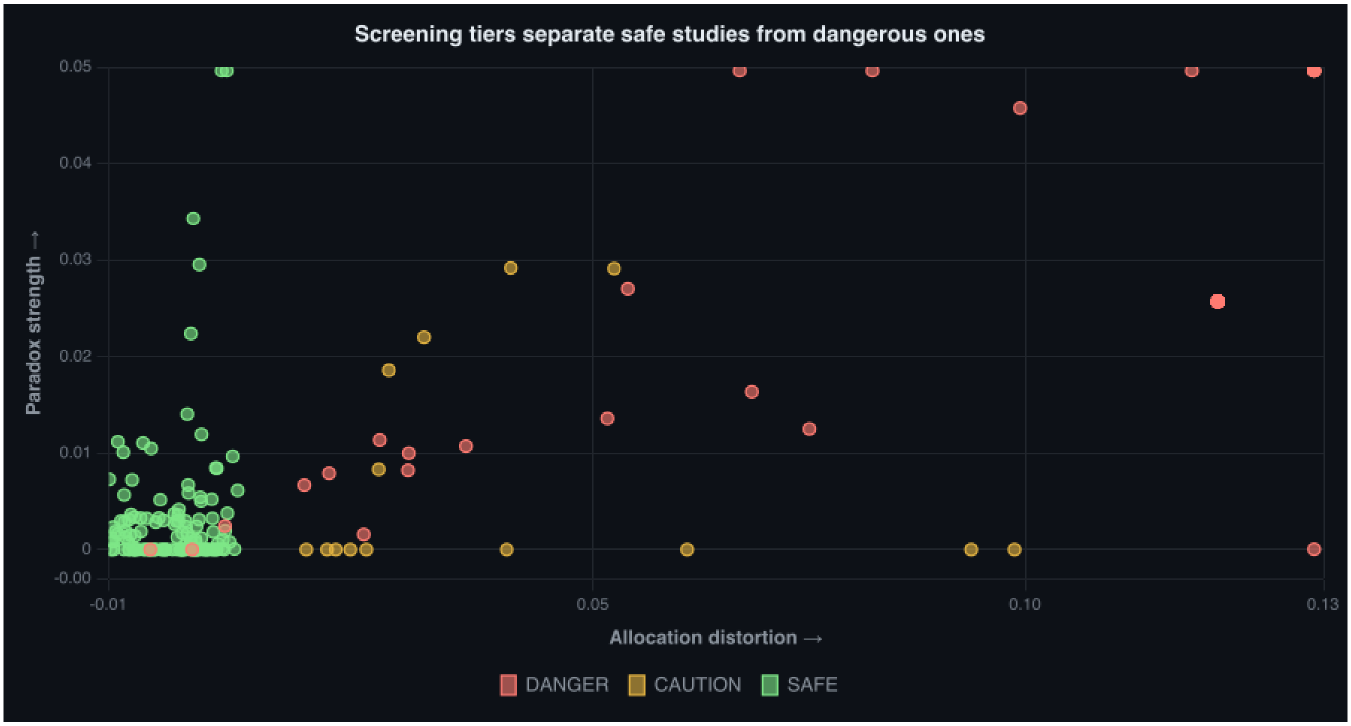
Loop replay: loop-85
Protecting invariants: 1

conc-36-perfect-balance-theorem · Proved (by construction)

Perfect allocation balance is a sufficient condition for no manifest Simpson's Paradox

Loop-86: inv-perfect-balance-no-flip PASS 118/118. All 118 TreatmentRankings with ArmSizeRatio=0.5 AND MaxStratumImbalance=0.0 have IsSignFlip=FALSE across all 8 domains and 19 confounder archetypes. Minimum imbalance at any sign-flip study: 0.033 (flu-vaccine-elderly-mortality, demographic-composition). Safety corridor [0.0, 0.033) contains zero sign-flips. This is a sufficient-condition theorem: perfect balance guarantees no manifest paradox, domain-agnostically.

Loop replay: loop-86
Protecting invariants: 1



conc-37-purity-inversion-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Purity inversion: high SignalPurity anti-predicts safe stratification — collider geometry maximizes it

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-87: collider/selection mean SignalPurity=0.916 (n=12); selection/frailty mean SignalPurity=1.000 (n=5) — the two mechanism classes that must NOT be stratified have the highest purity in the corpus. Demographic-composition (stratify-appropriate) mean SignalPurity=0.624 (n=35). inv-collider-purity-dominance PASS. H-purity-inversion confirmed. PurityTrapFlag encodes the trap for direct query. Researchers using data cleanliness as a quality heuristic will systematically misidentify collider/selection studies as...

Loop replay: loop-87
Protecting invariants: 1

Supporting findings (17)

Established in the instrument or observed in this corpus — one line each. Full evidence text is in the rulebook JSON (Conclusions table).

Established in instrument

conc-03-binary-excludes-partial · The unanimity criterion (IsReversal) incorrectly excludes Berkeley 1973 despite its being a Simpson-type paradox

conc-04-type-c-exists · Allocation can compress the pooled signal without flipping it (type-C): a class invisible to IsReversal and IsSignFlip but visible to AllocationDistortion

conc-05-real-data-generalizes · Two real published studies (reintjes-2000, radelet-1981) hydrate as type-A full reversals without schema changes

conc-06-policy-derivable · PolicyImplication is domain-neutral: the same formula produces researcher actions across medicine, law, and education

conc-07-model-self-portrait · ModelSummary witnesses the instrument's own epistemic coverage as a first-class derived fact

conc-11-reversal-recovery · The model can produce the allocation-corrected winner directly from existing DAG fields — no new data required

conc-15-latent-flip-potential · Type-D at observed allocation does not imply allocation-stable — 70% of SAFE studies flip under counterfactual reweighting
Sensitivity over sweep range f ∈ [0.05, 0.95]; not a claim about realistic reallocation alone.

conc-16-small-effect-fragile · In this corpus, allocation-stable Type-D rows carry larger observed pooled gaps than latent Type-D rows

conc-18-causal-role-flip-mode · Causal role predicts manifest vs latent reversal mode — confounders flip at observed allocation; collider/selection flips stay sweep-latent

conc-19-explained-tracks-confounding · IsParadoxExplained tracks adjust-resolvable confounding, not geometric distortion

conc-24-collider-no-manifest-v2 · Collider/selection studies produce zero manifest sign-flips at observed allocation — theorem candidacy

conc-25-cplus-magnitude-blindspot · Type-C+ carries the largest mean AllocationDistortion — invisible to is_sign_flip screens

conc-26-sweep-fragile-class · Sweep-fragile Type-D class: pooled-safe studies with wide counterfactual wander (≥4 in corpus)

conc-33-confounder-identity-layer · ConfounderIdentity ontology: cross-study accountable stratum archetypes (loop-80)

Scope / open question

conc-08-type-c-real · Does the published literature contain real type-C studies (compression without sign flip)?

Observed in this corpus

conc-01-paradox-is-derived · Simpson's Paradox is a derived fact from the DAG, not a modeled entity

conc-02-reversal-from-allocation · The reversal in kidney-1986 is entirely an allocation artifact — same rates with balanced allocation produce no reversal

Other findings, in brief (11)

Superseded corpus patterns, build/methodology synthesis notes, and candidate (not-yet-promoted) theorems — kept for the record, not central to the current story.

- conc-10-instrument-specifiable — The full instrument can be specified precisely enough for any researcher to apply it to a new study
- conc-14-scale-is-discovery-path — Cross-domain SignalPurity correlations require a pre-registered unbiased corpus of 50+ studies to distinguish pattern from selection artifact
- conc-17-economics-flip-free — Economics-domain 2×K encodings in this corpus produce zero sign-flips — composition studies read as Type D or C−
- conc-21-catalog-tags-not-predictors — Catalog ExpectedDistortionType tags are mechanism priors, not taxonomy predictors — exact match rate 19.4%
- conc-22-domain-flip-geometry — Domain flip-rate heterogeneity survives max_stratum_imbalance conditioning — economics 0%, epi/legal/sports ~27%
- conc-23-research-sweep-synthesis — Research sweep loops 62–66: confirmed laws vs corpus patterns vs deferred vocabulary
- conc-27-economics-encoding-selection — Economics Expected-A catalog tags mismatch observed Type D at 2×K encoding — selection not immunity
- conc-31-theorem-portfolio-synthesis — Four witnessed theorems: SignalPurity, CorrectedGap invariance, Explained↔Confounder, Collider-no-manifest — plus corpus patterns that remain non-theorems
- conc-32-expansion-wave-3-supersession — Expansion wave 3 (loop-77→78): six corpus patterns superseded at N=238 — theorems and invariants hold
- conc-34-latent-predominant-family — Latent-predominant mechanism family: selection-frailty, collider-proxy, and geographic-composition produce near-zero manifest flips
- conc-expansion-wave-synthesis — Expansion waves 1–2 (loops 68–72): adversarial stress tests survived; theorem candidacy patterns held or sharpened

Pre-registered discovery (loop-61)

Consistency checks — confirm implementation matches stated algebra (same tier as conc-12 / inv-signal-purity-sign-flip). Not independent discoveries about nature.

H-collider-no-manifest-theorem — PASS (consistency check)

At scale (≥10 collider/selection studies post expansion), zero manifest sign-flips: ColliderSelectionManifestCount = 0.

Expected: ColliderSelectionManifestCount = 0 AND ColliderSelectionCount >= 10

Observed: collN=11 collManifest=0

H-corrected-gap-invariant — PASS (consistency check)

CorrectedGap is allocation-invariant for every study: MAX(SweepStudySummary.SweepCorrectedGapRange) < 0.0001 and zero failing studies.

Expected: MaxStudySweepCorrectedGapRange < 0.0001 AND CorrectedGapInvariantFailCount = 0
Observed: maxRange=0.000000 fails=0

H-explained-bidirectional — PASS (consistency check)

IsParadoxExplained ⟺ confounder manifest sign-flip: ExplainedConfounderCount = ConfounderSignFlipCount, no false-positive explained rows, no unexplained confounder sign-flips.

Expected: ExplainedConfounderCount = ConfounderSignFlipCount AND FalsePositiveExplainedCount = 0 AND UnexplainedConfounderSignFlipCount = 0
Observed: explained=85 confFlip=85 falsePos=0 unexplained=0

H-identity-map-coverage — PASS (consistency check)

At least 95% of real-study StratumVariables carry a ConfounderIdentity map — ontology covers the corpus.

Expected: IdentityMapCoverageRate >= 0.95
Observed: IdentityMapCoverageRate=1.000000

H-perfect-balance-sufficient-safety — PASS (consistency check)

Perfect allocation balance (ArmSizeRatio=0.5 AND MaxStratumImbalance=0.0) is a domain-agnostic sufficient condition for no manifest Simpson's Paradox — no such study produces IsSignFlip=TRUE regardless of mechanism class, domain, or confounder identity.

Expected: PerfectBalanceFlipCount=0 AND PerfectBalanceStudyCount>=100
Observed: PerfectBalanceFlipCount=0 PerfectBalanceStudyCount=118

H-purity — PASS (consistency check)

Every sign-flip study has SignalPurity strictly below 0.5.

Expected: SignFlipSignalPurityMax < 0.5
Observed: maxPurity=0.499937

H-purity-inversion — PASS (consistency check)

Mechanism classes with PolicyDefault='do-not-condition' or 'do-not-naively-adjust' (collider-selection, selection-frailty) have strictly higher mean SignalPurity than all stratify-* mechanism classes. High SignalPurity anti-predicts safe stratification — it is a trap for researchers who use data cleanliness as a quality heuristic.

Expected: MeanPurityColliderSelection > MeanPurityAllStratifyClasses AND MeanPuritySelectionFrailty >= 0.95
Observed: MeanPurityDoNotCondition=0.908500 MeanPurityStratifyClasses=0.727900

H-theorem-portfolio — PASS (consistency check)

Four Category=theorem conclusions are witnessed: SignalPurity (conc-12/20) plus three new theorems from loops 73–75.

Expected: TheoremCount >= 4
Observed: theoremCount=8

H-unexplained-flips-only-nonconfounders — PASS (consistency check)

All unexplained sign-flips (IsSignFlip=TRUE AND IsParadoxExplained=FALSE) have ConfounderIdentity ∈ {id-contested-org-choice, id-collider-proxy} OR CausalRole=contested — no unexplained flip belongs to an unambiguous confounder identity.

Expected: UnexplainedConfounderSignFlipCount = 0
Observed: UnexplainedConfounderSignFlipCount=0

Corpus hypotheses — contingent patterns on this convenience sample. Directional inequalities only; not inferential. See conc-14.

H-age-identity-flip-rate — PASS (corpus hypothesis)

Age-composition identity (id-age-composition) produces manifest flip rate ≥ 40% among real studies — cross-domain demographic confounding is flip-prone, not latent-only.

Expected: AgeIdentityManifestFlipRate >= 0.40 AND AgeIdentityStudyCount >= 10
Observed: AgeIdentityManifestFlipRate=0.485714 AgeIdentityStudyCount=35

H-catalog-exact-match — PASS (corpus hypothesis)

Catalog ExpectedDistortionType exact-match rate is below 50% — curator tags behave as mechanism priors, not taxonomy predictors.

Expected: TypePredictionMatchRate < 0.5
Observed: exactRate=0.391489 (92/235)

H-causal-latent — PASS (corpus hypothesis)

Collider/selection studies produce latent-only flips under allocation sweep without manifest sign-flips at observed allocation.

Expected: ColliderSelectionManifestCount = 0 AND ColliderSelectionLatentOnlyCount >= 5
Observed: collManifest=0 collLatent=10 collN=11

H-causal-manifest — PASS (corpus hypothesis)

Confounder-labeled studies produce manifest sign-flips at observed allocation (IsSignFlip=TRUE) far more often than collider/selection studies — reversal mode differs by mechanism class.

Expected: ConfounderSignFlipCount > ColliderSelectionManifestCount AND ConfounderSignFlipCount >= 10
Observed: confFlip=85 collManifest=0

H-collider-identity-low-manifest — PASS (corpus hypothesis)

Collider-proxy identity (id-collider-proxy) produces ≤15% manifest flip rate — near-zero manifest, predominantly latent Type-D geometry.

Expected: ManifestFlipRate(cluster-id-collider-proxy) <= 0.15 AND StudyCount(cluster-id-collider-proxy) >= 5
Observed: ColliderIdentityManifestFlipRate=0.090909 ColliderIdentityStudyCount=11

H-collider-no-manifest-v2 — PASS (corpus hypothesis)

Collider and selection studies produce zero manifest sign-flips (DistortionType A or B) at observed allocation — including hernandez-diaz birth-weight collider tagged Expected A in catalog.

Expected: ColliderSelectionManifestCount = 0
Observed: collManifest=0

H-confounder-drift-type-d — FAIL (corpus hypothesis)

When studies are bucketed by real-world valid-time (cohort midpoint, not publication year), case-mix / geographic / situational confounder archetypes (MechanismClass in severity-case-mix, geography, sports-situation) show DriftDirection=rising toward clean Type-D outcomes in ConfounderDistortionTimeline, while age / institutional / SES archetypes (demographic-composition, organizational-unit, ses-income) show DriftDirection in (flat, single-decade) -- i.e. the drift is confounder-specific, not corpus-wide.

Expected: COUNTIFS(ConfounderDistortionTimeline[MechanismClass],"severity-case-mix|geography|sports-situation",ConfounderDistortionTimeline[DriftDirection],"rising") = COUNTIFS(ConfounderDistortionTimeline[MechanismClass],"severity-case-mix|geography|sports-situation") AND COUNTIFS(ConfounderDistortionTimeline[MechanismClass],"demographic-composition|organizational-unit|ses-income",ConfounderDistortionTimeline[DriftDirection],"rising") = 0
Observed: severity=falling geography=flat situational=rising age=falling institutional=rising socioeconomic=rising — expected severity/geography/situational in (rising,single-decade) and age/institutional/socioeconomic <> rising; institutional and socioeconomic both drifted rising, contradicting the hypothesis.

H-control-no-manifest-flip — PASS (corpus hypothesis)

Zero manifest sign flips among control studies. These studies were selected for absence of paradox, so no pooled-vs-stratified reversal should be present.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,Studies[IsControlStudy],TRUE) = 0

Observed: ControlManifestFlipCount=0 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-control-safety-corridor — PASS (corpus hypothesis)

No control study with MaxStratumImbalance < 0.033 (the safe-corridor threshold) has IsSignFlip=TRUE. The safety corridor hypothesis predicts zero flips below this boundary.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,TreatmentRankings[MaxStratumImbalance],"<0.033",Studies[IsControlStudy],TRUE) = 0

Observed: ControlSafeCorridorFlipCount=0 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-control-type-d-predominance — PASS (corpus hypothesis)

At least 70% of control studies encode as Type-D (neutral allocation, no sign flip) — Type-D is the expected basin for studies selected for absence of paradox.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,Studies[IsControlStudy],TRUE) / COUNTIF(Studies[IsControlStudy],TRUE) < 0.3

Observed: ControlFlipRate=0.000000 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-cplus-magnitude — PASS (corpus hypothesis)

C+ studies carry higher mean AllocationDistortion than C- studies and Type-D studies — magnitude inflation is the largest quantitative error class and invisible to is_sign_flip.

Expected: CPlusAvgDistortion > CMinusAvgDistortion AND CPlusAvgDistortion > TypeDAvgDistortion

Observed: C+=0.111288 C-=0.029823 D=0.000329

H-explained-confounder — PASS (corpus hypothesis)

IsParadoxExplained is TRUE for all and only confounder sign-flip studies — adjust-resolvable confounding is the unique condition that makes a reversal fully explained.

Expected: ExplainedConfounderCount = SignFlipCount AND ExplainedNonConfounderCount = 0

Observed: ExplainedConfounderCount=85 ConfounderSignFlipCount=85

H-geographic-stable-type-d — PASS (corpus hypothesis)

Geographic-composition identity (id-geographic-composition) has ≥25% Type-D fraction — stable latent-D presence distinct from manifest-flip-prone identities.

Expected: TypeDFraction(cluster-id-geographic-composition) >= 0.25 AND StudyCount(cluster-id-geographic-composition) >= 3

Observed: GeographicTypeDFraction=0.647059 GeographicStudyCount=17

H-latent-d — PASS (corpus hypothesis)

More than half of Type-D (SAFE) studies would flip their pooled winner under counterfactual allocation reweighting.

Expected: LatentTypeDFraction > 0.5

Observed: fraction=0.811594

H-mechanism-other-drain — PASS (corpus hypothesis)

After ontology drain, id-mechanism-other contains ≤5 studies — the catch-all is no longer the dominant bucket.

Expected: StudyCount(cluster-id-mechanism-other) <= 5

Observed: MechanismOtherStudyCount=0

H-purity-trap-flag-coverage — PASS (corpus hypothesis)

PurityTrapFlag=TRUE covers ≥80% of collider/selection studies and ≤5% of demographic-composition confounder studies — the flag correctly discriminates the trap from the safe-stratification zone.

Expected: PurityTrapCoverage(collider-selection) >= 0.80 AND PurityTrapFalsePositiveRate(demographic-composition) <= 0.05

Observed: ColliderSelectionCoverage=0.818200 DemographicFalsePositiveRate=0.000000

H-safety-corridor-lower-bound — PASS (corpus hypothesis)

The minimum observed MaxStratumImbalance among all manifest sign-flip studies is 0.033 (flu-vaccine-elderly-mortality). No sign-flip occurs below this threshold — the gap [0.0, 0.033) is a corpus-level safety corridor.

Expected: MinSignFlipStratumImbalance >= 0.033 AND SafeCorridorFlipCount=0

Observed: MinSignFlipStratumImbalance=0.033024 SafeCorridorFlipCount=0

H-selection-frailty-zero-manifest — PASS (corpus hypothesis)

Selection-frailty identity (id-selection-frailty) produces zero manifest sign-flips — pure latent-D geometry, no allocation-reversal at observed weights.

Expected: ManifestFlipCount(cluster-id-selection-frailty) = 0 AND StudyCount(cluster-id-selection-frailty) >= 3

Observed: SelectionFrailtyManifestFlipCount=0 SelectionFrailtyStudyCount=5

H-severity-epi-latent-only — PASS (corpus hypothesis)

Disease-severity identity within epidemiology domain shows zero manifest flips — latent-only geometry in that domain.

Expected: ManifestFlipCount(cell-id-disease-severity-x-epidemiology) = 0

Observed: SeverityEpiManifestFlipCount=0

H-severity-medicine-manifest — PASS (corpus hypothesis)

Disease-severity identity within medicine domain has manifest flip rate ≥45% — severity drives reversals in clinical settings.

Expected: ManifestFlipRate(cell-id-disease-severity-x-medicine) >= 0.45

Observed: SeverityMedicineManifestFlipRate=0.533333

H-severity-vs-age-latent — FAIL (corpus hypothesis)

Disease-severity identity shows lower Type-D latent fraction than age-composition identity — severity drives manifest reversals more than hidden sweep sensitivity.

Expected: SeverityIdentityLatentFractionAmongTypeD < AgeIdentityLatentFractionAmongTypeD

Observed: SeverityIdentityLatentFractionAmongTypeD=0.000000 AgeIdentityLatentFractionAmongTypeD=0.000000

H-small-effect — PASS (corpus hypothesis)

Allocation-stable Type-D studies carry larger observed pooled gaps than latent Type-D studies.

Expected: AvgPooledGapStableD > AvgPooledGapLatentD

Observed: stable=0.113368 latent=0.020205

H-synthetic-edge-coverage — PASS (corpus hypothesis)

All four corpus theorem boundary regions (near-zero-distortion, equal-weight sign-flip, zero-corrected-gap, collider-purity) are covered by at least one synthetic study.

Expected: COUNTIF(Studies[IsSynthetic],TRUE) >= 4 AND all four boundary regions represented

Observed: SyntheticStudyCount=9 near-zero-distortion=t equal-weight-sign-flip=t zero-corrected-gap=t collider-purity=t

H-ultra-fragile — PASS (corpus hypothesis)

At least four Type-D studies are sweep-fragile: signal_purity=1, allocation_distortion=0, sweep_pooled_gap_range>0.3, allocation_fragility>=10 — perfectly safe by pooled metrics yet latent-flip under reweighting.

Expected: SweepFragileCount >= 4
Observed: SweepFragileCount=8

H-unexplained-nonconfounder — PASS (corpus hypothesis)

Studies with IsSignFlip=TRUE and CausalRole != confounder (contested or mediator) have IsParadoxExplained=FALSE.

Expected: ContestedOrMediatorExplainedCount = 0
Observed: ContestedOrMediatorExplainedCount=0

Invariant health

Critical invariant failures: 0
Definitional (cannot fail by construction — see definitions above): 22
Judgment-crossing (touch hand/LLM-encoded CausalRole): 4
Vacuous (0 pass / 0 fail — empty universe, proves nothing yet): 0

All 26 critical invariants pass — 22 are transpiler regression checks against stated algebra (cannot fail by construction), 4 cross into the hand/LLM-encoded CausalRole field, and 0 are currently vacuous (empty universe). See "How to read this report" above.

Warning-tier geometry checks currently failing (5) — not critical, tracked separately, listed for completeness:

- inv-import-session-ready: Loop-67+: expansion backlog loaded; bulk encode session ready when >=20 candidates and >=15 encode-ready. (fail=1)
- inv-pooled-gap-crosses-zero: Legacy sweep check from ratio-based Type A (pre-loop-70). Type A is now unanimity-based; cross-zero is informative but not definitional. (fail=1)
- inv-pooled-gap-wanders: Pooled gap should move under allocation reweighting for reversal studies. Warning-only: small-magnitude reversals may have narrow sweep range. (fail=1)
- inv-type-a-ratio-negative: Legacy ratio geometry check (pre-loop-70). Type A is now IsStratumUnanimous; ratio band retained as warning-only witness. (fail=4)
- inv-type-b-ratio-lt-neg1: Legacy ratio geometry check (pre-loop-70). Type B is now heterogeneous stratum directions; ratio threshold retained as warning-only witness. (fail=2)

Most misleading studies (by paradox strength)

COVID-19 CFR: Italy vs China by Age (von Kügelgen 2021) — type C+, strength 0.2743, distortion 0.529, purity 3.6% (CAUTION)
Triathlon On-Time Finish: Sex by Age Cohort — type A, strength 0.1455, distortion 0.245, purity 28.9% (DANGER)
Smoking and Mortality by Age (Cochran / Surgeon General 1964) — type A, strength 0.1091, distortion 0.167, purity 25.7% (DANGER)
UC Berkeley Graduate Admissions (Bickel et al. 1975) — contested confounder — type B, strength 0.0969, distortion 0.193, purity 24.9% (DANGER)
UC Berkeley Admissions: Six Largest Departments (Bickel et al. 1975) — type B, strength 0.0944, distortion 0.184, purity 18.8% (DANGER)
Exercise and Cholesterol Control by Age Group (Framingham-style) — type D, strength 0.0800, distortion 0.008, purity 95.2% (SAFE)
Baseball Batting Averages: Lefebvre vs Fairly World Series (Day 1994 / baseball paradox canon) — type A, strength 0.0606, distortion 0.082, purity 20.9% (DANGER)
Whickham Smoking Cohort — 20-year Mortality by Smoking Status (Appleton et al. 1996) — type B, strength 0.0538, distortion 0.119, purity 26.9% (DANGER)

Rulebook & repo (full derivation)

Formulas, Loops build history, InvariantChecks, and Conclusions rows live in the rulebook JSON. README.md carries the full framing.

Project folder: https://github.com/EffortlessAPI/effortless-rulebooks/tree/main/rulebook-examples/simpsons-paradox
README (full repo guide): README.md

Rulebook SSoT (formulas, Loops, InvariantChecks, Conclusions): effortless-rulebook/simpsons-paradox-rulebook.json

Rulespeak (plain-English business rules): rulespeak/rulespeak.md

Postgres substrate (views, init-db.sh): effortless-postgres/README.md

Standalone HTML explorer: simpsons-paradox-explorer.html

Live API (with ./start.sh): GET localhost:3001/api/invariant-checks · /api/treatment-rankings · /api/model-summary